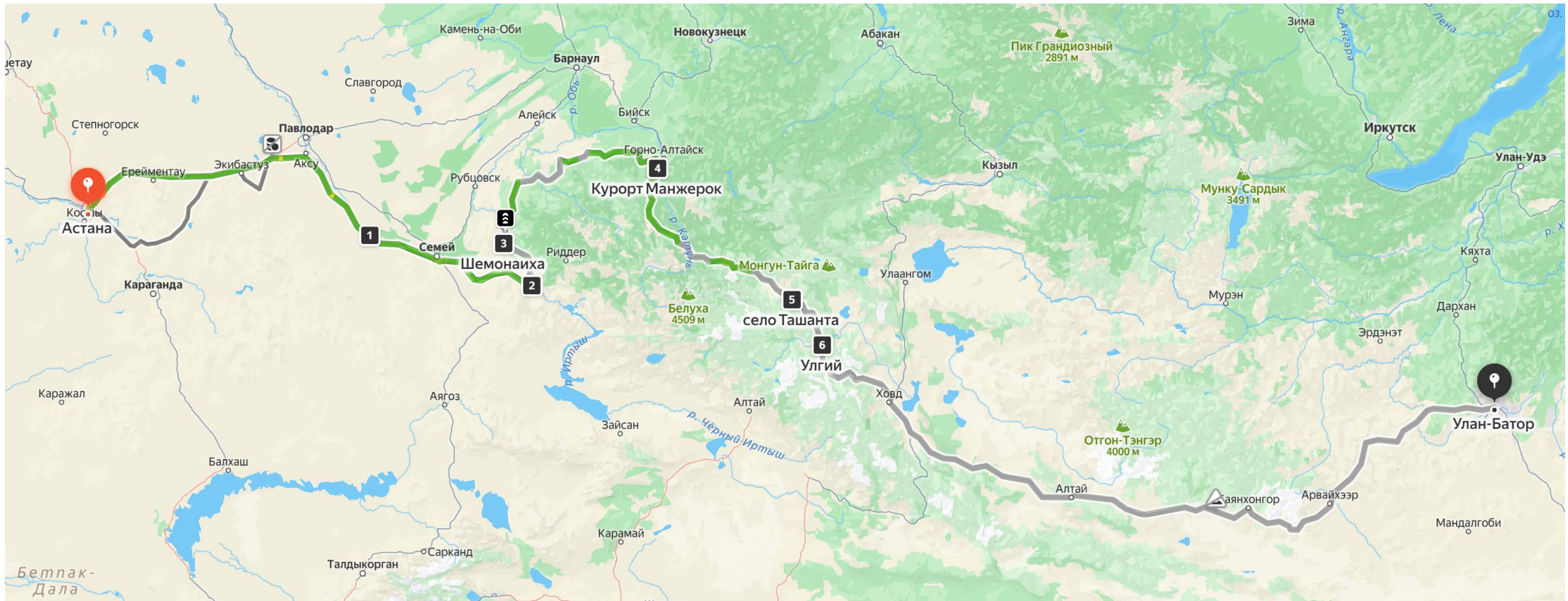


I'm motivated to take part in the conference.

Kazakhstan -> Russia -> Mongolia, 6 days (19-24), 3800 km, and endless delight



About us

Association of Remote Sensing Companies in Kazakhstan



- Founded in 2020;
- Office in Astana, Kazakhstan;
- Includes more than 30 members;
- Protect the business interests of members;
- Authorized in more than 5 ministries of Kazakhstan;
- Attended in development of 5 laws, related to geospatial market.

EOSpatial LLP

EOSPATIAL

- Founded in 2018;
- Office in Astana, Kazakhstan;
- Official partner of 7 satellite operators;
- Doing business in Central Asia countries (Kazakhstan, Uzbekistan, Kyrgyzstan)



Where we start?



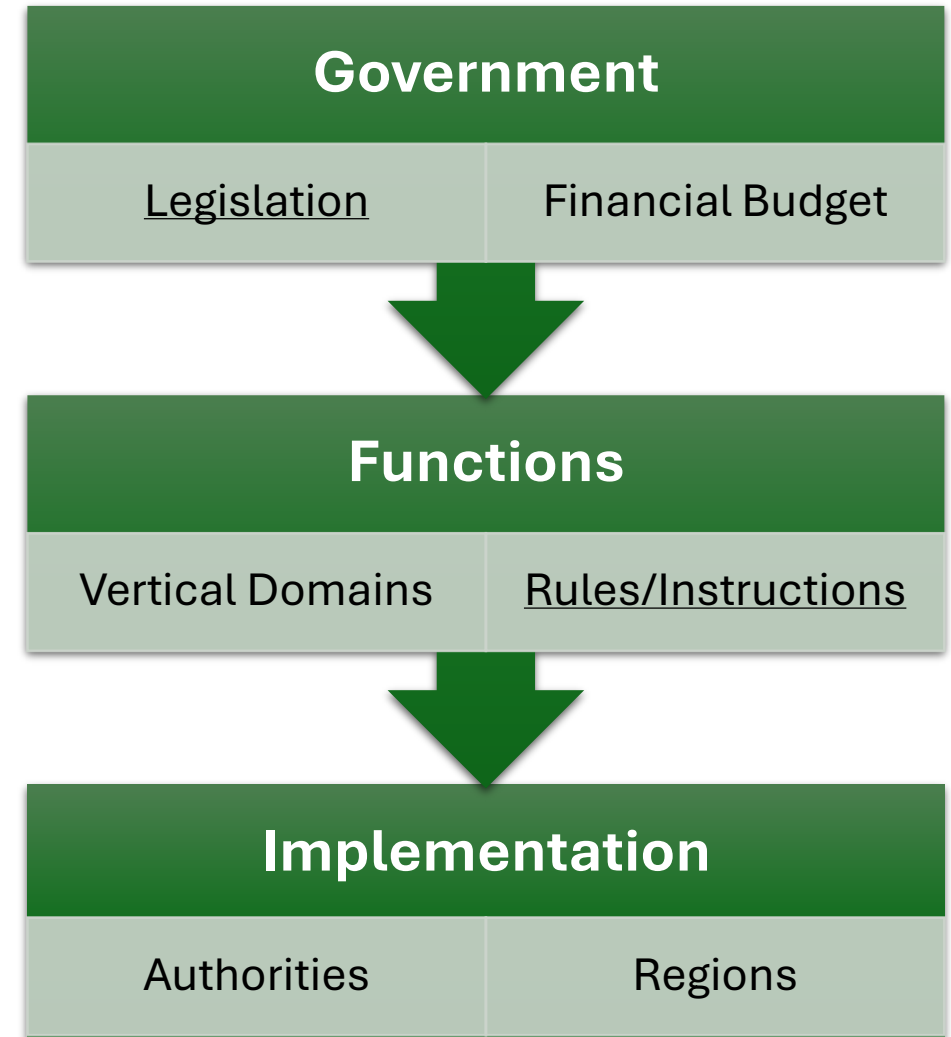
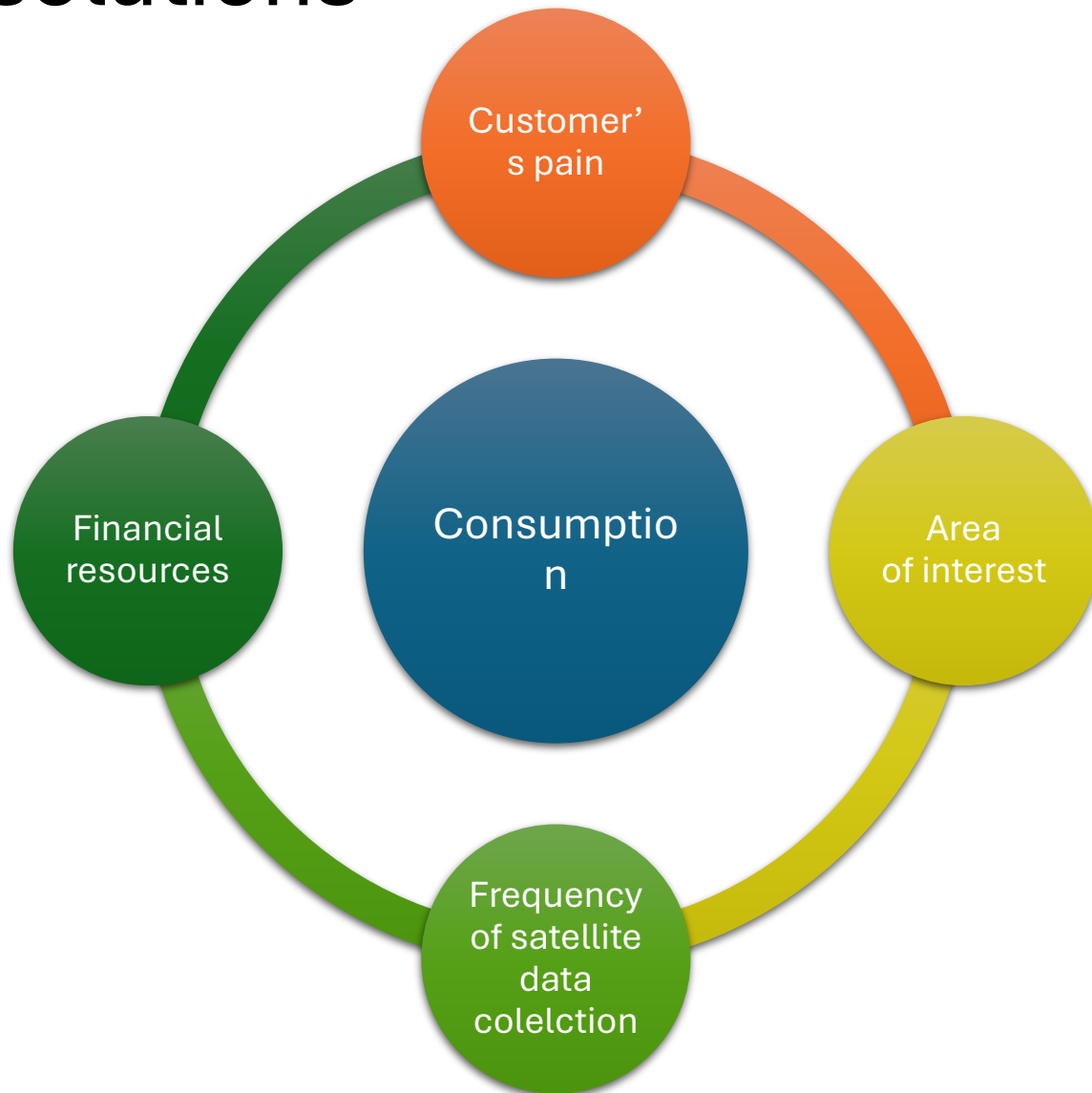
*"Kazakhstan is ready to **support Mongolia's plans to create a remote sensing system for the Earth.** We could also train qualified specialists for this industry. Our country places special emphasis on digital development."*

April 21, 2026

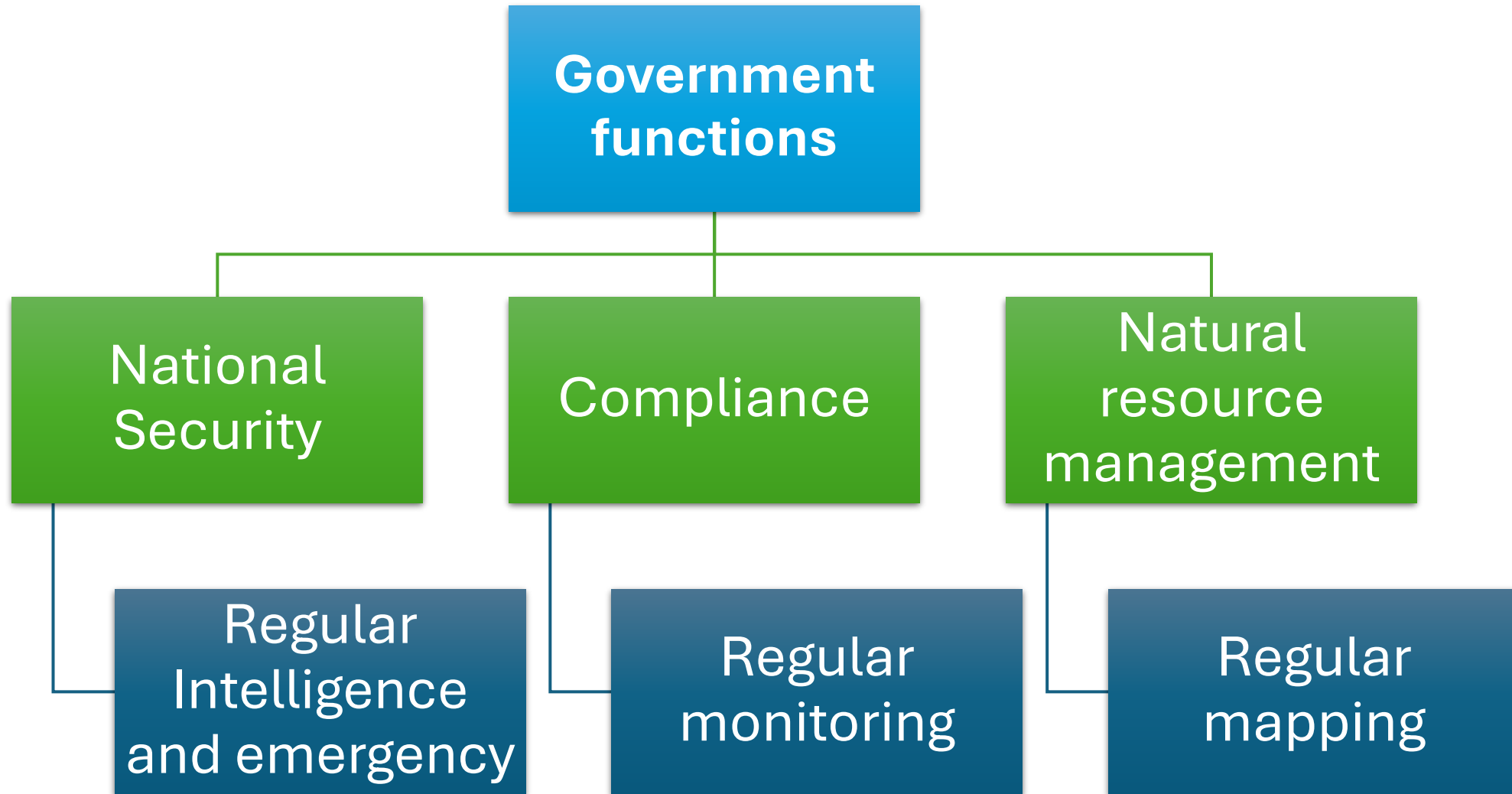
Source:

<https://kapital.kz/economic/147213/kazakhstan-gotov-pomoch-mongolii-sozdat-sistemu-distancionnogo-zondirovaniya-zemli.html>

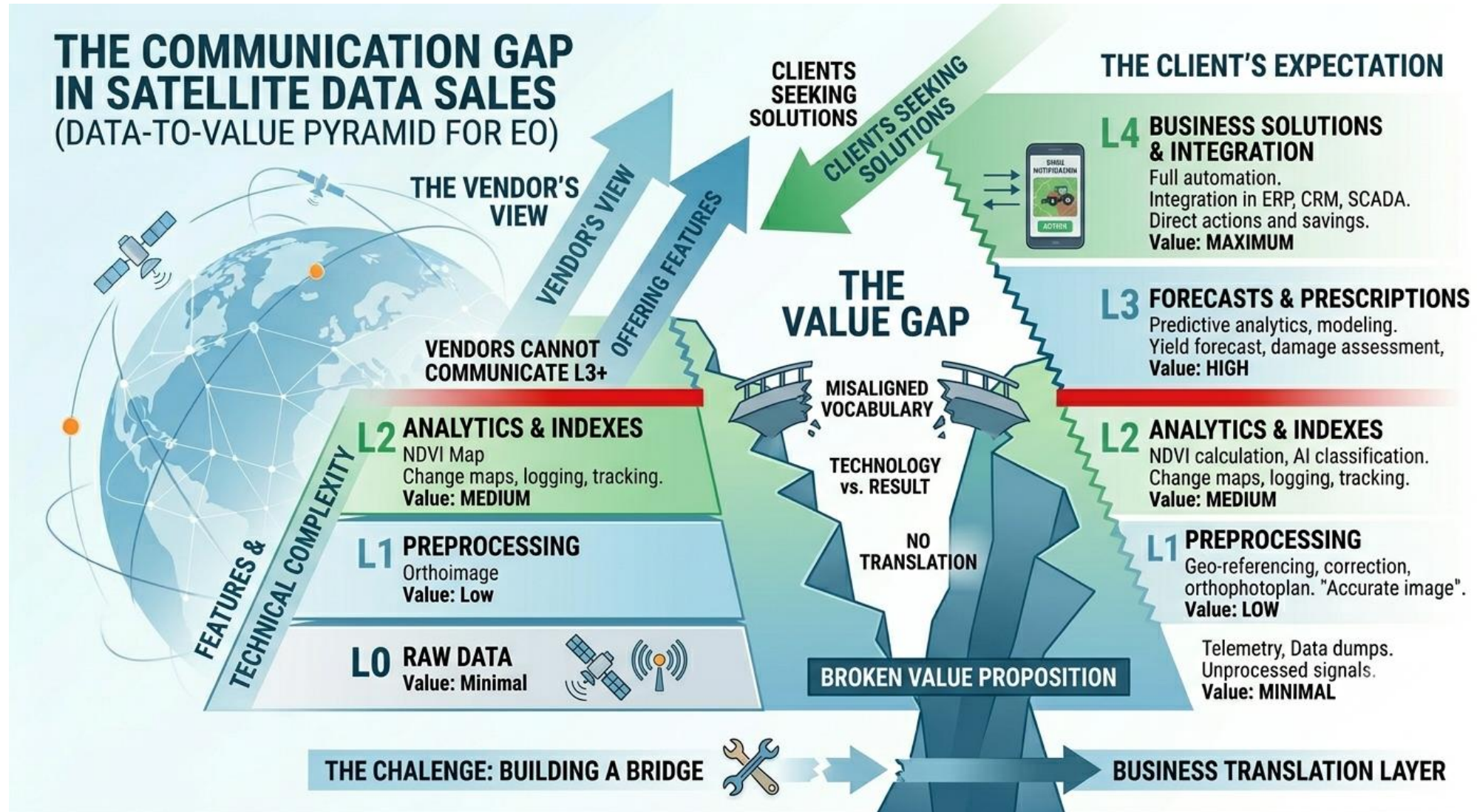
The Government is the best consumer for EO solutions



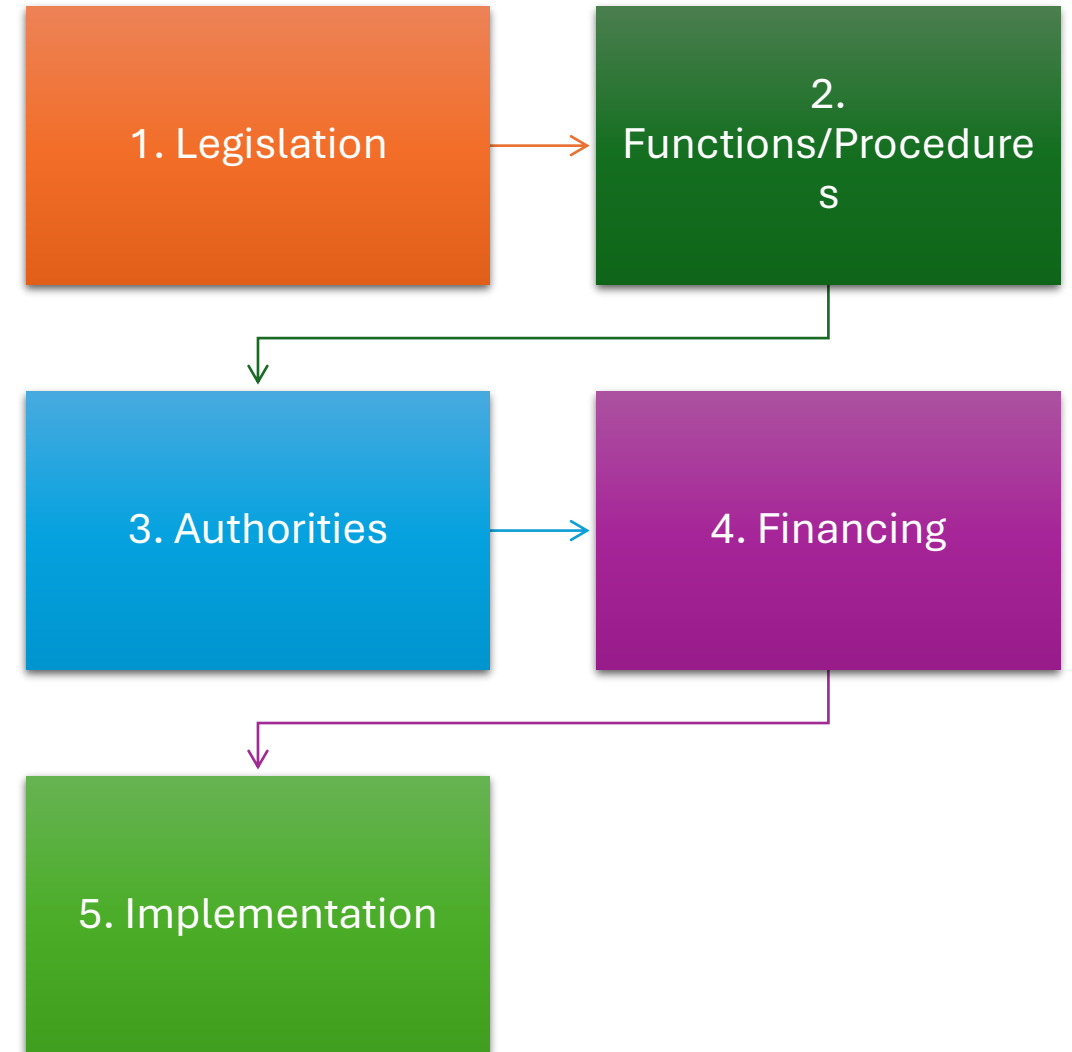
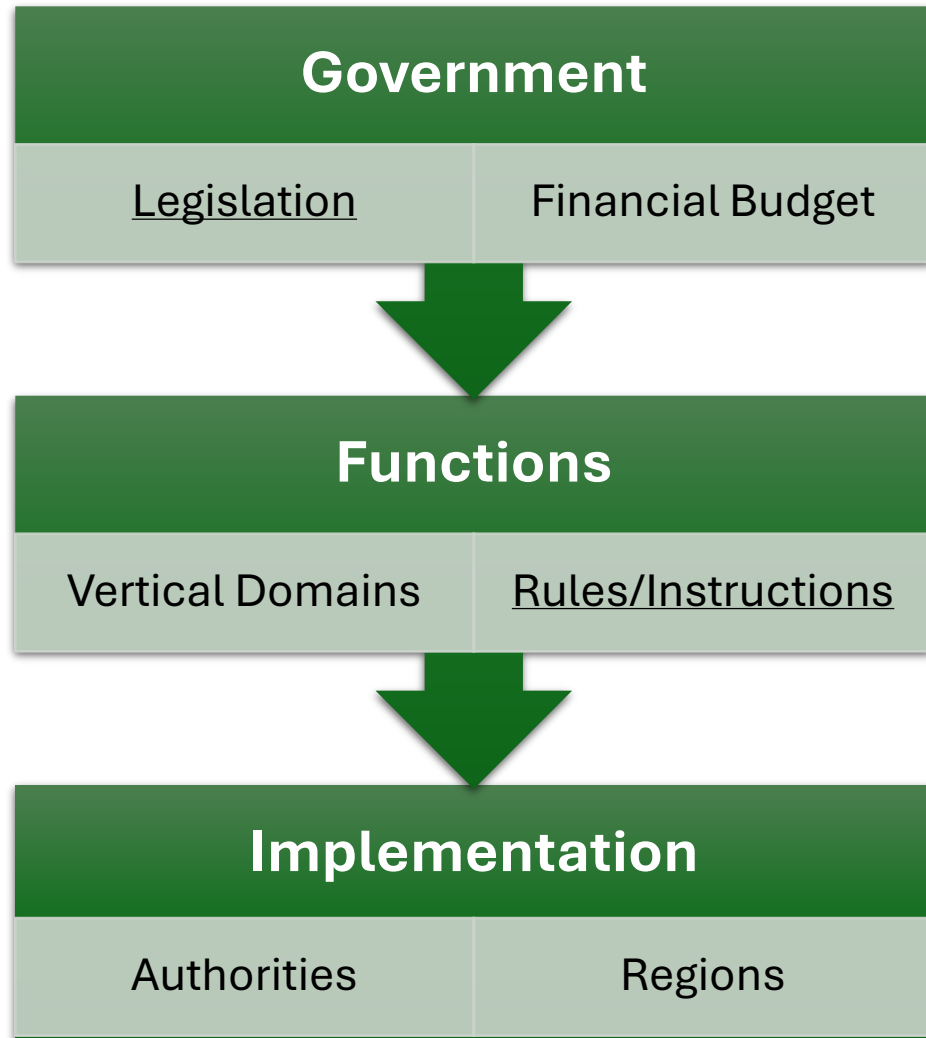
The most interesting government's functions for implementation of satellite-based EO technologies



Satellite earth observation. Pixels or Value?



Satellite earth observation. Pixels or Value?



Where we start?



Clause 3 of Article 6 of the Constitution of the Republic of Kazakhstan:
"The land and its subsoil, water, flora and fauna, and other natural **resources belong to the people. The state exercises ownership rights on behalf of the people.** Land may also be privately owned on the grounds, conditions, and within the limits established by law."



Prerequisites for the widespread use of satellite-based monitoring for compliance of natural resource usage

Geographic conditions

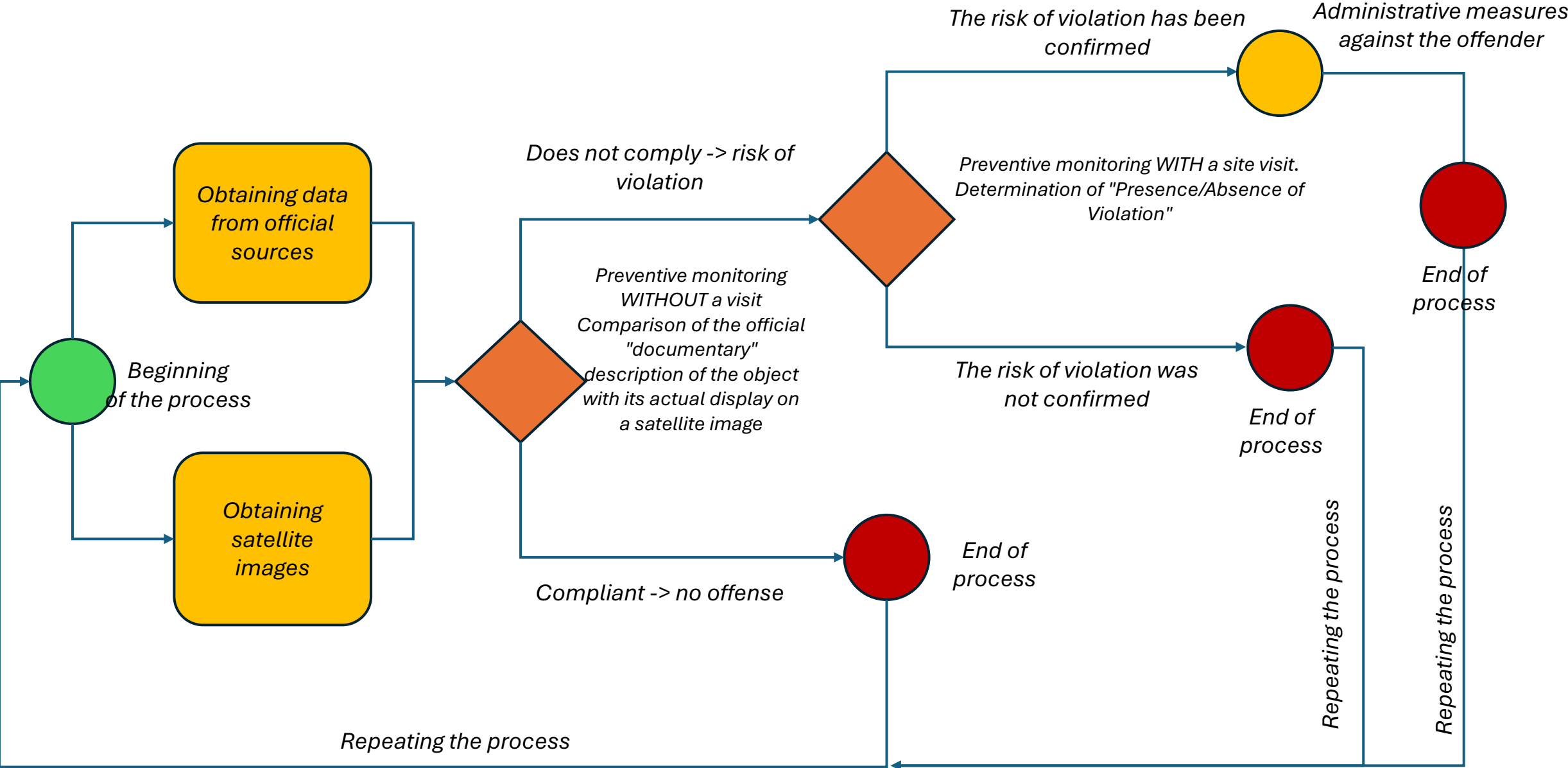
- The country's vast area (2.7 million square kilometers);
- Low population density (7.39 people/square kilometer);
- A large number of monitoring facilities;
- Long distances between settlements and to natural resource management facilities;

Economic conditions

- Limited government resources for oversight (staff, transport, fuel)
- Low communication/internet coverage;
- Low level of digitalization and use of objective oversight systems;
- Corruption potential in the sphere of government oversight;



Business process of satellite based monitoring for compliance



Compliance Sample #1



Песчаный карьер, река Нура, Акмолинская область.
Водоохранная зона/наличие лицензии на добычу ОПИ



Compliance Sample #2



Н.п. Акши, обл. Жетысу, оз. Акколь.
Водоохранная зона/градостроительное
планирование/застройка/земельные отношения

Compliance Sample #3



Актыбинская область.

Рекультивация участков добычи полезных ископаемых

An example of the systemic effect of the implementation of satellite-based monitoring for compliance

2019

"Identifying land that is being used inefficiently is extremely difficult, so we couldn't make hasty decisions on such a sensitive issue as land use rights. At the same time, the need to implement modern monitoring methods based on the latest technologies has long been urgent in the area of state control over agricultural land use. The general public, including members of the National Commission, noted the need to transition to space monitoring to identify and remove unused and inefficiently used land."



2022

Amendments to the Land Code of the Republic of Kazakhstan. Inclusion of Article 145-1 "Procedure for Conducting Preventive Inspections Without Visiting the Subject (Object) of Inspection."

The use of Earth remote sensing data is being introduced to promptly prevent and address the non-use or irrational use of agricultural land.



2025

The Ministry of Artificial Intelligence noted that space monitoring is an important tool for land resource management. According to the data presented, work has been underway in three areas since 2018: crop production, pastures, and land use. As an example, they cited figures for 2024, which indicate that **approximately 2 million hectares of unused or partially used arable land have been identified.**



The values of Satellite-based monitoring for compliance

To improve the efficiency of the state control/monitoring system

- Establishment of a risk management system (RMS) identifying entities/activities that contain indicators of violations;
- Rational planning of limited resources for more effective implementation of state control/monitoring;
- Strengthening the evidence base for administrative action against violators;
- Ability to cover larger territories/facilities with the monitoring system;

To improve the functioning of the public administration system

- Strengthening the digitalization policy of the public administration system;
- Eliminating the "administrative" monopoly and technical bottleneck in collecting "unconfirmed" data within state control/monitoring;
- Strengthening objectivity over subjectivity/additional impetus for "data-driven government";
- Further balancing of the internal system of checks and balances between control/monitoring participants;
- Reducing the scope for corruption;

Legislative norms governing the application of satellited-based monitoring results for **compliance**;

Land Code of the Republic of Kazakhstan

•Article 145-1. "3. Preventive control without visiting the subject (object) of control is carried out on the basis of remote sensing data of the Earth and identification of farm animals on agricultural lands based on the facts of their non-use for their intended purpose and (or) irrational use."

Code "On Subsoil and Subsoil Use"

Article 66. "2. ... Monitoring of the fulfillment by subsoil users of their obligations under a contract (license) for subsoil use shall be carried out by the competent authority (the state body that is a party to the contract and (or) that has issued the license for subsoil use) through the analysis of reports submitted by subsoil users in accordance with this Code, space monitoring data containing signs of violations of the legislation of the Republic of Kazakhstan, and information received from other sources in accordance with the legislation of the Republic of Kazakhstan."

Forest Code of the Republic of Kazakhstan

Article 19-1. "9. . . The basis for appointing an inspection is an order from the first head of the authorized agency or the first head of the territorial subdivision. The grounds for issuing the relevant order are: 4) space monitoring data containing signs of violations of the legislation of the Republic of Kazakhstan."

Water Code of the Republic of Kazakhstan

Article 53. "3. To conduct an inspection, basin water inspectorates shall first conduct a predictive assessment of water bodies that potentially pose high risks of unauthorized water use and non-compliance with the prohibitions on economic activity on water bodies, in water protection zones, and strips established by this Code, based on: 3) remote sensing of the Earth."

Article 56. "2. Preventive monitoring without a visit shall be conducted by analyzing: 3) remote sensing data."

Legislative norms governing the application of satellite based monitoring of **environment**;

Земельный Кодекс РК

•**Статья 159.** «1. Мониторинг земель представляет собой систему базовых (исходных), оперативных, периодических наблюдений за качественным и количественным состоянием земельного фонда, в том числе с **использованием данных дистанционного зондирования Земли**, проводимых в целях государственного контроля за использованием и охраной земель, своевременного выявления происходящих изменений, их оценки, прогноза дальнейшего развития и выработки рекомендаций по предупреждению и устранению последствий негативных процессов»

Кодекс «О недрах и недропользовании»

Статья 73. «1. Государственный мониторинг недр представляет собой систему наблюдений за состоянием недр, в том числе **с использованием данных дистанционного зондирования Земли**, для обеспечения рационального управления государственным фондом недр и своевременного выявления их изменений, оценки, предупреждения и устранения последствий негативных процессов.»

Лесной Кодекс РК

Статья 61. «Государственный мониторинг лесов представляет собой систему наблюдений, оценки и прогноза состояния и динамики лесного фонда, в том числе **с использованием данных дистанционного зондирования Земли**, в целях государственного управления в области охраны, защиты, пользования лесным фондом и воспроизводства лесов, сохранения биологического разнообразия и экологических функций лесов.....»

Водный Кодекс РК

Статья 36. «5. Источниками информации для государственного мониторинга водных объектов и водных ресурсов служат: 3) результаты дистанционного зондирования Земли»

Legislative norms governing the application of satellite based monitoring for **emergency cases**;

Law of the Republic of Kazakhstan "On Civil Defense"

- Article 41. "2. General civil defense measures to prevent emergency situations include:
- **1) the organization of monitoring systems, including the use of remote sensing**, civil defense alerts, and the protection of territories and facilities from emergency situations."



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Head of the council of the Association of RS companies / CEO

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